INCH-POUND
A-A-59983
12 June 2017

COMMERCIAL ITEM DESCRIPTION

TRANSIT LEVEL, AUTO

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. The automatic transit level provides accurate measurements of horizontal and vertical angles with azimuth and elevation gradations in relative and/or absolute scales of degrees. This CID lists the minimum performance, design, functional, environmental, reliability requirements, and quality assurance provisions for a transit level, auto.

2. CLASSIFICATION. The transit level covered by this CID shall be commercially-available equipment that has factory-installed options and may be modified to the extent necessary to meet the following description. The test set shall be class 3, in accordance with MIL-PRF-28800.

2.1 Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. SALIENT CHARACTERISTICS

3.1 Item description. The automatic transit level is a precision optical instrument used to measure horizontal angles and vertical levels with azimuth and elevation gradations in relative and/or absolute scales. These angle measurements are used to determine linear distances in the vertical and/or horizontal planes by means of triangulation or absolute magnitude to assist in the positioning of markers, signs, antennas, illuminators, reflectors, optical systems, sights, earth/soil grading, and machinery positioning for test, measurement, and diagnostic support equipment (TMDE) support for the development, repair, and maintenance of avionics, microwave, navigational aids, radar, satellite, radio communications equipment, and other applications in civil, electrical, and mechanical engineering.

3.2 Display information. The transit level shall provide level and angle reading in degrees. Reticle and stadia lines/marks shall be visible through the optical viewer/sight.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: WR-AFMC 404 SCMS/GUEEB, 235 Byron St., Robins AFB, GA 31098 or email SPEC99@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

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3.3 Magnification (optical power). The transit level shall be capable of providing a minimum optical magnification of 30 times magnification (30X). See table I.

3.4 Field of view. The transit level shall have a minimum effective field of view of ≥1 ft 6 in @ 100 ft (0.45 m @ 30.48 m) compliant with ISO 17123, parts 1 thru 3. See table I.

3.5 Barrel and pincushion (spherical) distortion. The transit level shall have a maximum barrel and pincushion (spherical) distortion of 3% at the shortest and longest focal distance in accordance with ISO 17123, parts 1 thru 3.

3.6 Transmittance of light. The transit level shall have a maximum transmitted light attenuation of 35% between 495 nm and 570 nm.

3.7 Ultra-violet (UV) radiation (A and B). The transit level shall have a minimum transmitted UV irradiation attenuation of 98% at a wavelength of 400 nm and shorter from a clear sky source in accordance with ISO 21348 as described in ISO 15469. The irradiation intensity at the ocular lens or view finder shall be less than the 25 mW/m2 (25µW/cm2) from a clear sky source per McKinlay-Diffey weighted UV irradiance in accordance with TD-No. 95, 625, and 921.

3.8 Angular accuracy. The transit level shall have a minimum effective angular accuracy of ±0.5° in compliance with ISO 17123, parts 1 thru 3. See table II.

3.9 Horizontal axis (azimuth) rotation accuracy. The transit level horizontal-axis rotation accuracy variance shall be less than 1° ±0.5° displayed in compliance with ISO 17123, parts 1 thru 3. See table I.

TABLE I. Functional parameters.

Function Description	Parameters
Effective Diameter of Objective Lens	≥40 mm (1.57 in)
Minimum Magnification	≥30X
Azimuth Rotation Accuracy ∡°	1° ±0.5°
Leveling Accuracy Minimum @ 100 ft (30.48 m)	±1/4 in @ 100ft (±6.35 mm @ 30.48 m)
Image Display	Erect
Field of View Minimum	≥1 ft 6 in @ 100 ft (0.45 m @ 30.48 m)
Stadia Ratio Multiplier Constant	1:100
Stadia Additive Constant	0
UV Ray Filter Blocking (Lens Coating)	UV_A & UV_B ≥ 98%
Ingress Protection Level	IP54

3.10 Horizontal axis (azimuth) rotation range. The transit level horizontal-axis range shall have a minimum angle sweep of ±360°.

3.11 Horizontal axis (azimuth) attitude. The transit level shall have a circular vial “Bulls-Eye Bubble” or two tubular vials aligned at 90° from each other on the horizontal plane.

3.12 Horizontal axis rotation clamp. The transit level shall have a horizontal (azimuth) adjustment knob to set and restrict motion on the horizontal plane.

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3.13 Vertical axis (elevation angle) range. The transit level vertical-axis range shall have two stadia line/mark gradations for 1:100 stadia ratio for distance estimation to align the pole, staff, or optical rod.

3.14 Vertical axis leveling accuracy. The transit level minimum leveling accuracy at 100 ft (30.48 m) shall be ±1/4 in (±6.35 mm) per table II.

TABLE II. Resolution accuracy range in D.M.S.

Accuracy in inches @ 100 feet.	Accuracy in cm @ 30.48 meters	Angle Degrees ∡°	D.M.S
1.0000	2.53999	0.0477464488	0° 2’ 51.88”
0.7500	1.90500	0.0358098645	0° 2’ 8.90”
0.5000	1.27000	0.0238732422	0° 1’ 25.90”
0.2500	0.63500	0.0119366208	0° 0’ 42.97”
0.1250	0.31750	0.0059683104	0° 0’ 21.48”
0.0625	0.15875	0.0029841552	0° 0’ 10.74”
0.0465	0.11821	0.0022222222	0° 0’ 8.00”
0.0407	0.10343	0.0019444444	0° 0’ 7.00”

3.15 Design and construction. The transit level shall be used in both built-up and rough-terrain environments. The transit level shall be designed and constructed in accordance with all applicable European Union (EU) requirements with the “CE” marking affixed, International Organization for Standardization (ISO), the International Electro-technical Commission (IEC), the International Telecommunication Union (ITU), and the American National Standards Institute (ANSI). The transit level shall be compliant with the Restriction of Use of Hazardous Substances (RoHS) and Waste from Electrical and Electronic Equipment (WEEE) guidelines and policies. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.16 Ingress protection (dust and water). The transit level shall have a minimum ingress-protection rating of IP54 in accordance with IEC 60529 for water and dust within the temperature operating range.

3.17 Metal coupling. The transit level shall have a Tribrach and tripod base plate assembly construction that minimizes corrosion and galvanic reactions on contact surfaces.

3.18 Rotation couplers. The transit level shall have a continuous or smooth rotational movement that is detent free between the transit head and the Tribrach and tripod base plate assembly.

3.19 Rotational force. The transit level shall have a continuous or smooth rotational movement in which the force needed to rotate the transit level head is less than 1.5 inch-pounds.

3.20 Level indicators. The transit level shall have an obstruction-free view to the circular and tubular vials (bubble levels) while in a standard operating configuration. The transit level shall have an internal magnetic/mechanical compensator and compensator lock.

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3.21 Level indicator viewfinders. The transit level shall have an external prism lens aligned with the ocular lens on either side for viewing the circular (bubble or bulls-eye) level while in a standard operating configuration.

3.22 Tripod coupler. The transit level shall have a threaded, mechanical screw on coupler with a five eights of an inch dash eleven (5/8”-11) UNC-thread bolt hole that holds and secures the transit level Tribrach base plate assembly to the tripod or any other supporting platform.

3.23 Tripod unit. The transit level shall be issued with a tripod that has a threaded, mechanical coupler with a five eights of an inch dash eleven (5/8”-11) UNC-thread bolt hole that holds and secures the transit level. The tripod shall be a flat head unit with an extended height of 60 in ±4 in (152 cm ±10 cm), and a closed or collapsed height of 36 in ±4 in (91.4 cm ±10 cm). It shall have a quick-release clamp and wing screws.

3.24 Tripod rod. The transit level shall be issued with a surveyor’s “rod” or “staff” that is rugged, with a rectangular or "C" channel-shaped, leveling-grade shape with an abrasion-resistant and corrosion-resistant coating made of non-conductive material. It shall have a minimum extended length of 10 ft (3.048 meters), and locking telescoping sections. The collapsed length shall be between 48 in and 60 in (121.9 to 152.4 cm).

3.25 Operating temperature range. The transit level shall meet the specifications listed herein with full capabilities within the temperature range of 14 to 131 °F (-10 to 55 °C).

3.26 Storage temperature range. The transit level shall meet or exceed manufacturer’s specifications for stored equipment of 14 to 131 °F (-10 to 55 °C).

3.27 Heat dissipation and ventilation. Heat exposure and absorption shall not alter angular measurements within standard operating parameters. Heat dissipation shall be sufficient to allow for prevention, buildup, and retention of moisture, or cause erratic measurements.

3.28 Altitude requirements. The transit level shall meet its operating performance and accuracy while operating at a minimum elevation of 12,000 ft (3,658 meters).

3.29 Calibration and maintenance adjustments. No special tools, software, technical manuals, or documentation shall be required for the alignment, adjustment, repair, and calibration of the transit level, except those provided with the transit level. A software license shall be issued for each physical instrument when the system is designed for computer-based calibration. Any special or unique accessories not common to a secondary calibration laboratory required for calibration of this item shall be furnished by the manufacturer, to include, but not limited to, software, fixtures, tools, and standards.

3.30 Calibration period. The design of the transit level shall provide readily-accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test-set case or modules. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours, whichever occurs first. At the end of this interval, 85% of the units shall remain within calibration tolerances in accordance with manufacturer’s specifications.

3.31 Reliability. The design of the transit level shall be such that under normal use and operation, the mean time before failure (MTBF) exceeds one year or 2,080 hours of operation with a statistical certainty equal to or greater than 95% based on a 40-hour week and 52-week year.

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3.32 Maintainability. The transit level shall meet the maintainability criterion in accordance with MIL STD 1472, paragraphs 5.9 through 5.9.18, and 5.9.

3.33 Preventive maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800, paragraphs 3.14.2 and 3.14.3.

3.34 Corrective maintenance. The transit level shall be designed to meet reparability criterion of the manual-of-field procedures for testing geodetic and surveying instruments, per ISO 17123-3.

3.35 Dimensions. The transit level shall occupy less than 0.75 ft^3, including the Tribarch.

3.36 Weight. The total weight of the transit level shall, excluding accessories and manuals, comply with MIL STD 1472, paragraph 5.9.11.3.3, for weight and size restrictions for single-person lift & handling, and shall not exceed a one-person lift of 25 pounds (11.4 kilograms).

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials or biobased content to the maximum extent practicable, in accordance with Federal Acquisition Regulation, Part 23.403.

5. PRODUCT CONFORMANCE. Verification of the delivered transit level will be performed at the delivery point by the end user.

6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Source of documents.

7.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks listed below form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-28800	-	General Specification for Test Equipment for use with Electrical and Electronic Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1472	-	Human Engineering

(Copies of these documents are available at https://assist.dla.mil.)

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7.1.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529	-	Degrees of Protection Provided by Enclosures (IP Code) - Edition 2.2 Consolidated Re-print

(Copies of these documents are available at http://www.iec.ch/emc/basic_emc/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 15469	-	Spatial distribution of daylight— CIE standard general sky.
ISO 17123-1	-	Optics and Optical Instruments - Field Procedures for Testing Geodetic and Surveying Instruments - Part 1: Theory.
ISO 17123-2	-	Optics and Optical Instruments - Field Procedures for Testing Geodetic and Surveying Instruments - Part 2: Levels.
ISO 17123-3	-	Optics and Optical Instruments - Field Procedures for Testing Geodetic and Surveying Instruments - Part 3: Theodolites.
ISO 21348	-	Space environment (natural and artificial) - Process for determining solar irradiances.

(Copies of these documents are available at http://www.iso.org/.)

INTERNATIONAL COMMISSION ON ILLUMINATION (CIE)

CIE S 011:2003	-	CIE Standard Overcast Sky and Clear Sky

(Copies of these documents are available online at http://www.cie.co.at/ or from CIE, Babenbergerstraße 9/9A, A-1010 Vienna, Austria.)

WORLD METEOROLOGICAL ORGANIZATION – GLOBAL ATMOSPHERE WATCH

TD-No. 95	-	Field Workshop on Intercalibration of Conventional and Remote-Sensed Sea Surface Temperature Data
TD-No. 625	-	Report of the WMO Meeting of Experts on UV-B Measurements, Data Quality and Standardization of UV Indices

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TD-No. 921	-	Report of the WMO-WHO Meeting of Experts on Standardization of UV Indices and their Dissemination to the Public

(Copies of these documents are available online at http://www.wmo.int/pages/prog/arep/gaw/gaw_home_en.html or from the World Meteorological Organization/Global Atmosphere Watch, 7bis, avenue de la Paix, Case postale No. 2300, CH-1211 Geneva 2, Switzerland.)

7.2 Ordering data. Procurement documents must specify: title, number, date of this CID, the issue of the ASSIST Online, https://assist.dla.mil to be used in the solicitation, and preservation and packaging requirements.

7.3 Key words.

Calibration

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MILITARY INTERESTS

Custodian:
Air Force – 84
Army – MI

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

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